Variable dictionary of all my nodes that describe my Bayesian network for determining if a child who presents to hospital with suspected pneumonia has a bacterial or viral pneumonia, or no pneumonia.

Node	Description	States	Parents	Relationships To Parents, Justify arcs
Pneumonia	Represents the type of pneumonia a child may have: bacterial, viral, or no pneumonia.	Bacterial, Viral, No Pneumonia	Age, Vaccine Doses	1. Age -> Pneumonia Since younger children are more vulnerable to pneumonia than older children, age has a direct causal relationship with pneumonia. Younger age groups may have weaker immune systems, making them more susceptible to respiratory infections. 2. Vaccine Doses -> Pneumonia Fully vaccinated children are less likely to contract pneumonia, indicating that vaccine doses serve as a protective factor. Thus, vaccine doses can be
				considered a parent variable of pneumonia, as they influence the likelihood of developing the condition.
Age	Represents the age of the child, which can influence the likelihood of pneumonia.	Under 6 months, 6 months to 2 years, Above two years	-	Age has no parents in this context because it is a foundational factor. It serves as an independent variable that can influence various health outcomes, including pneumonia.
Fever	Represents whether the child has a fever, a common sign of bacterial pneumonia.	High fever, Low fever, No fever	Pneumonia	Pneumonia-> Fever Pneumonia often leads to fever as a response to infection. When the body detects the presence of pathogens causing pneumonia, it activates the immune response, which frequently results in an elevated body temperature (fever). This relationship shows pneumonia is a parent for fever, as having pneumonia increases the likelihood of developing a fever.

Rapid Breathing	Represents whether the child is experiencing rapid breathing, a sign of bacterial pneumonia.	True, False	Pneumonia, Age	1. Pneumonia -> Rapid Breathing Pneumonia can lead to rapid breathing (tachypnea) as the body attempts to compensate for reduced oxygen exchange in the lungs. This relationship indicates that pneumonia is a parent influencing the occurrence of rapid breathing. 2. Age -> Rapid Breathing Age can influence the likelihood of rapid breathing, especially in young children and the elderly. Younger children may have a naturally higher respiratory rate compared to older children and adults due to their smaller lung capacity and higher metabolic rates. This is why age is parent of rapid breathing.
Crackling Sound	Represents whether there is a crackling sound in the lungs, often observed with a stethoscope in bacterial pneumonia.	True, False	Pneumonia	Pneumonia -> Crackling Sound Pneumonia can cause a crackling or "rales" sound in the lungs due to the presence of fluid, mucus, or inflammation in the air sacs (alveoli). This relationship shows that pneumonia is a parent variable that influences the presence of abnormal lung sounds.
X-ray Result	Represents the results of an X-ray test, which can help diagnose pneumonia (e.g., consolidation, pleural effusion, or normal).	Normal, Consolidation with PE, Consolidation without PE	Pneumonia	Pneumonia -> X-Ray result The presence of pneumonia affects the results of an X-ray by leading to characteristic changes in the lung's appearance, such as consolidation, or consolidation with PE. This is why Pneumonia is parent of X-ray result.
Vaccine Doses	Represents the number of pneumococcal vaccine doses the child has received, which protects against bacterial pneumonia.	Zero, One, Two, Three	-	Vaccine Doses has no parents because it is an independent factor, since the number of vaccine a child receives is not affected by any nodes in the network.

Updating my structure from the previous simple Bayesian network, adding in parameters(parameterizing and adding the CPTs) after collecting relevant information from medical literature.

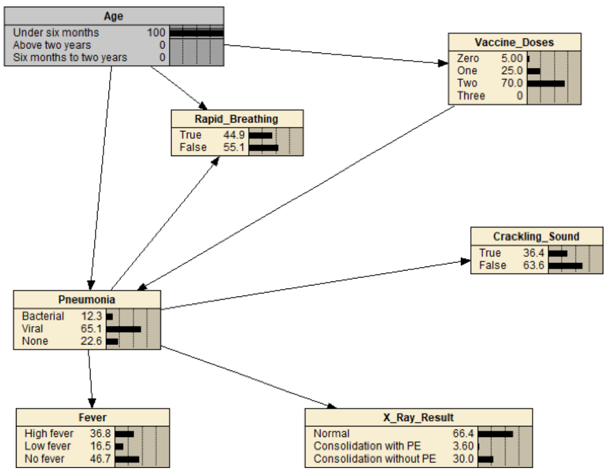
Node	Description	States	Parents	Relationships To Parents, Justification and source of Parameters and estimates
Pneumonia	Represents the type of pneumonia a child may have: bacterial, viral, or no pneumonia.	Bacterial, Viral, No Pneumonia	Age, Vaccine Doses	1. Age -> Pneumonia Since younger children are more vulnerable to pneumonia than older children, age has a direct causal relationship with pneumonia. Younger age groups may have weaker immune systems, making them more susceptible to respiratory infections. 2. Vaccine Doses -> Pneumonia Fully vaccinated children are less likely to contract pneumonia, indicating that vaccine doses serve as a protective factor. Thus,
				vaccine doses can be considered a parent variable of pneumonia, as they influence the likelihood of developing the condition. The CPT of pneumonia is built from sources ME and LIT.
Age	Represents the age of the child, which can influence the likelihood of pneumonia.	Under 6 months, 6 months to 2 years, Above two years	-	Age has no parents in this context because it is a foundational factor. It serves as an independent variable that can influence various health outcomes, including pneumonia. No prior distribution needed since the child's age is known (ME).

Fever	Represents whether the child has a fever, a common sign of bacterial pneumonia.	High fever, Low fever, No fever	Pneumonia	Pneumonia-> Fever Pneumonia often leads to fever as a response to infection. When the body detects the presence of pathogens causing pneumonia, it activates the immune response, which frequently results in an elevated body temperature (fever). This relationship shows pneumonia is a parent for fever, as having pneumonia increases the likelihood of developing a fever. The likelihood of high or low fever varies with the type of pneumonia. High fever is more common in bacterial pneumonia.(ME), Source of parameters taken from ME.
Rapid Breathing	Represents whether the child is experiencing rapid breathing, a sign of bacterial pneumonia.	True, False	Pneumonia, Age	1. Pneumonia -> Rapid Breathing Pneumonia can lead to rapid breathing (tachypnea) as the body attempts to compensate for reduced oxygen exchange in the lungs. This relationship indicates that pneumonia is a parent influencing the occurrence of rapid breathing. 2. Age -> Rapid Breathing Age can influence the likelihood of rapid breathing, especially in young children and the elderly. Younger children may have a naturally higher respiratory rate compared to older children and adults due to their
				smaller lung capacity and higher metabolic rates. This is why age is parent of rapid breathing. The type of pneumonia and age affects the likelihood of having rapid breathing(LIT). Source of parameters taken from ME.

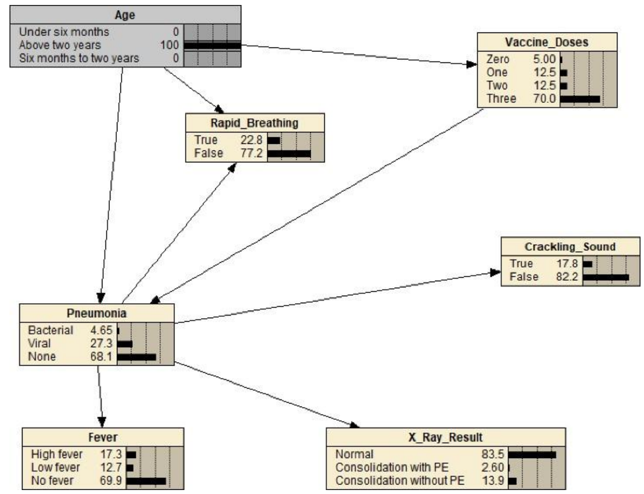
Crackling Sound	Represents whether there is a crackling sound in the lungs, often observed with a stethoscope in bacterial pneumonia.	True, False	Pneumonia	Pneumonia -> Crackling Sound Pneumonia can cause a crackling or "rales" sound in the lungs due to the presence of fluid, mucus, or inflammation in the air sacs (alveoli). This relationship shows that pneumonia is a parent variable that influences the presence of abnormal lung sounds. Crackling sounds are more common with bacterial pneumonia (ME). Rare in viral pneumonia or no pneumonia (ME). Source of parameters: ME.
X-ray Result	Represents the results of an X-ray test, which can help diagnose pneumonia (e.g., consolidation, pleural effusion, or normal).	Normal, Consolidation with PE, Consolidation without PE	Pneumonia	Pneumonia -> X-Ray result The presence of pneumonia affects the results of an X-ray by leading to characteristic changes in the lung's appearance, such as consolidation, or consolidation with PE. This is why Pneumonia is parent of X-ray result. The type of pneumonia affects likelihood of consolidation appearing on an X-Ray(LIT). Source of parameters: LIT
Vaccine Doses	Represents the number of pneumococcal vaccine doses the child has received, which protects against bacterial pneumonia.	Zero, One, Two, Three	Age	Age-> Vaccine Doses Vaccine doses are affected by age.(Eg, if the child is below 6 months, its impossible to be fully vaccinated, my probabilities reflect this). Source of parameters: ME

Example of how my parameterized Bayesian network works:

When a 5 month old girl and a 3 year old boy is presented to hospital with suspected pneumonia at about the same time.   Without finding out any of the patient details, symptoms, or test results:



5 month old girl is present to hospital

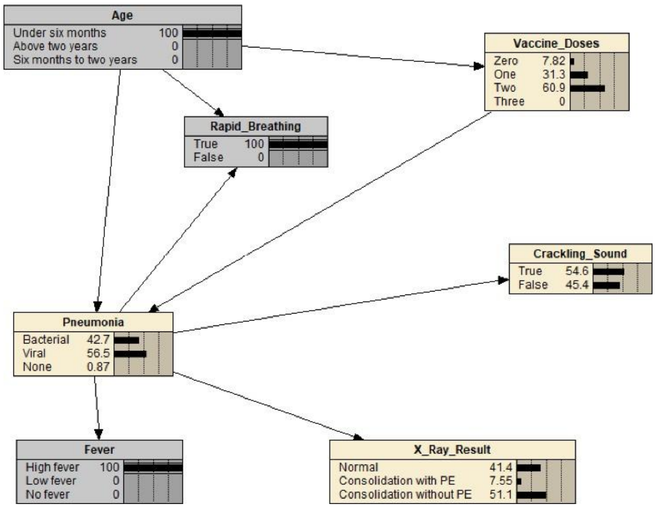


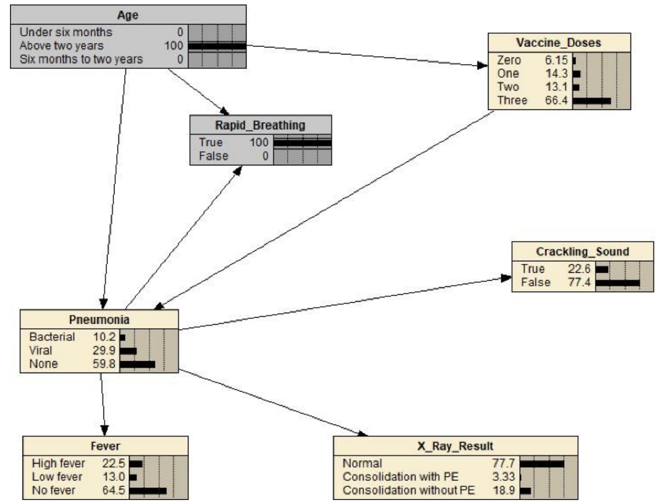
3 year old boy is present to hospital

The 5 month old girl has a 12.3% chance of getting bacterial pneumonia, the 3 year old boy has a 4.65%chance of getting bacterial pneumonia. This means that the girl is 7.65% more likely to have bacterial pneumonia than the boy. The girl has a 12.3 + 65.1 = 77.4% chance of getting any type of pneumonia.

Continuing on from the previous example:

Suppose both children present with rapid breathing. The girl also has a high fever, while we are not yet aware of whether the boy has a fever.





5 month old girl with rapid breathing and high fever

3 year old boy with rapid breathing

The 5 month girl is more likely to have bacterial pneumonia. The girl has a 42.7% probability of getting bacterial pneumonia, whereas the boy has a 10.2% probability of getting bacterial pneumonia. The girl has a 0.87% probability of no pneumonia, whereas the boy has a 59.8% probability of no pneumonia.

Node: X_Ray_Result | Apply | OK

Chance | % Probability | Reset | Close

Pneumonia	Normal	Consolidation with PE	Consolidation without PE
Bacterial	5	15	80
Viral	68	2	30
None	95	2	3

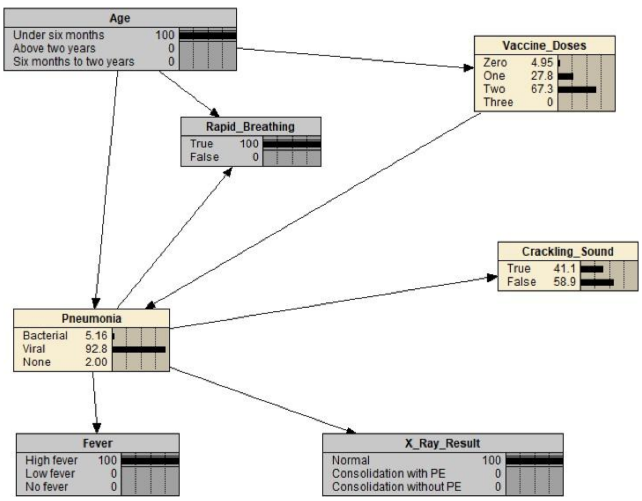
Looking at the X-ray CPT, the true positive rate for bacterial pneumonia is 80 + 15 = 95%,

The true positive rate for viral pneumonia is 2 + 30 = 32%.

Looking at the X-ray CPT, the overall false positive rate 2+3= 5%.

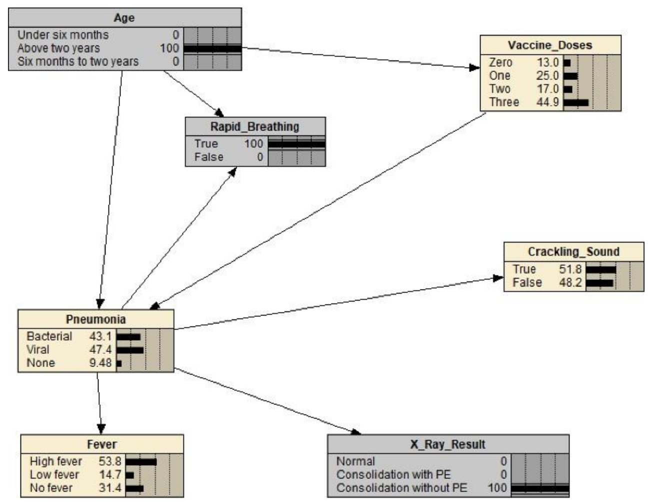
The true positive rate for any pneumonia or the false positive rate for bacterial/viral pneumonia cannot be directly calculated from the X-ray CPT alone because the CPT provides conditional probabilities for X-ray results given the type of pneumonia (bacterial, viral, or no pneumonia). However, it doesn't provide the base rates or prior probabilities for how often each type of pneumonia occurs in the population, nor does it account for the overall prevalence of any pneumonia.

When we let both children have an X-ray. The boy's X-ray shows consolidation, while the girl's looks normal.



5 month old girl with rapid breathing,

high fever and normal X-ray result

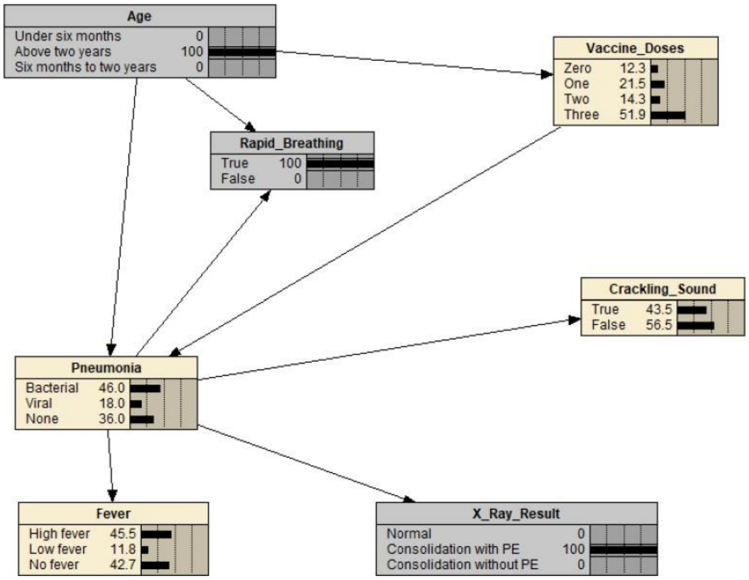


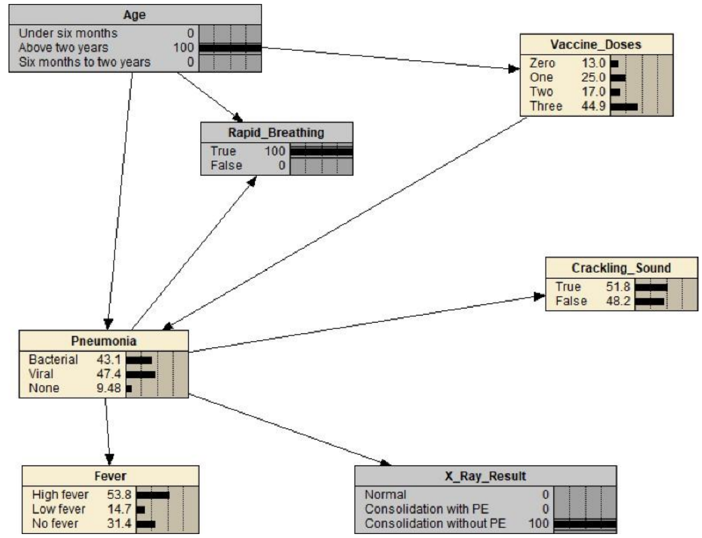
3 year old boy with rapid breathing,

and X-ray result showing consolidation

The 3 year old boy is more likely to have bacterial pneumonia. The probability that the girl has no pneumonia is 2.00%, while the probability that the boy has no pneumonia is 9.48%.

If instead, the boy's X-ray shows consolidation, while the girl's looks normal:





3 year old boy with rapid breathing,

and X-ray result of consolidation with PE

3 year old boy with rapid breathing,

and X-ray result of consolidation without

PE.

If the boy's X-ray result shows consolidation with PE, the probability of him having bacterial pneumonia is 46%. The probability increased from when the boy's X-ray result shows consolidation without PE, with probability of having bacterial pneumonia being 43.1percent. The probability increased because, according to LIT, the study shown is that , on quote 'bacterial pneumonia is quite likely to lead to consolidation appearing on an X-ray, with an estimated 80% showing evidence of consolidation, compared to 30% for viral pneumonia. An extra 15% for bacterial pneumonia (and 2% for viral pneumonia) will also show evidence of a PE.' This means that PE is more commonly associated with bacterial pneumonia than viral pneumonia. So, the presence of consolidation with PE strongly indicates bacterial pneumonia as 15% of bacterial pneumonia cases show PE, compared to only 2% of viral pneumonia cases. When PE does appear with consolidation, it is a strong diagnostic indicator of bacterial pneumonia. Therefore, if the boy's X-ray had shown both consolidation and PE, the probability that he has bacterial pneumonia would increase because the combination of those two findings is far more indicative of bacterial than viral pneumonia.

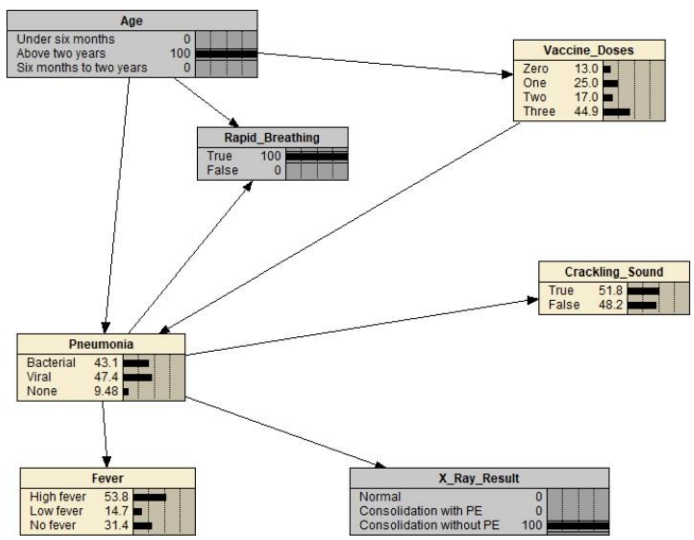
From previous analysis, we can see that, if the x ray shows consolidation, or consolidation with PE, the chances of him having bacterial pneumonia is higher. This uses information from LIT, which states that there is an estimated 80% showing evidence of

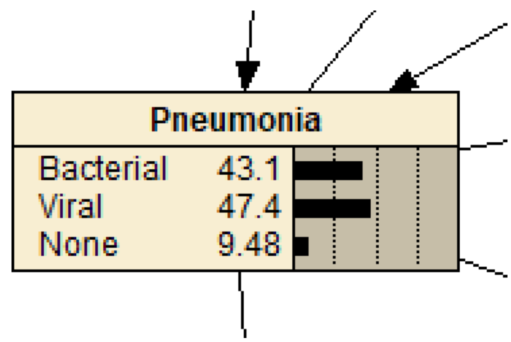
Bacterial pneumonia, with an extra 15% for bacterial pneumonia will show evidence of PE. The poster probabilities of bacterial pneumonia also rely on information from ME, in that the risk of bacterial pneumonia is higher for newborns and younger children, and especially in those not vaccinated.

The posteriors are sensitive to the parameters used in the model:

For the boy(Xray shows consolidation, girl's xray looks normal),

using my original parameters:





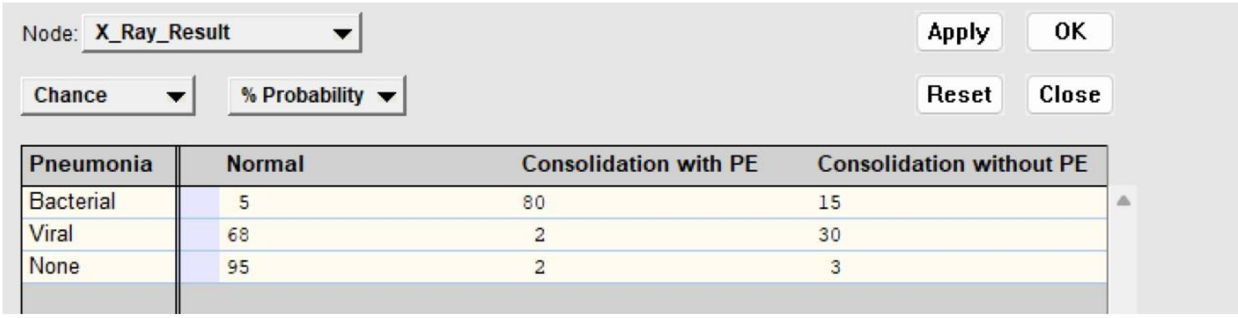
The probability of the boy having bacterial pneumonia is 43.1%.

Originally, we can see that the CPT of X_Ray_Result node is like so.

Node: X_Ray_Result | Apply | OK
Chance | % Probability | Reset | Close

Pneumonia	Normal	Consolidation with PE	Consolidation without PE
Bacterial	5	15	80
Viral	68	2	30
None	95	2	3

Then, I change the values of the CPT like so:


Node: X_Ray_Result

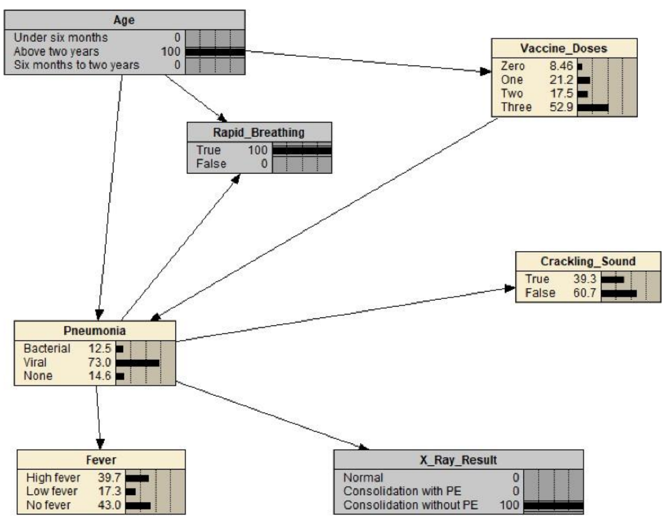
Chance | % Probability

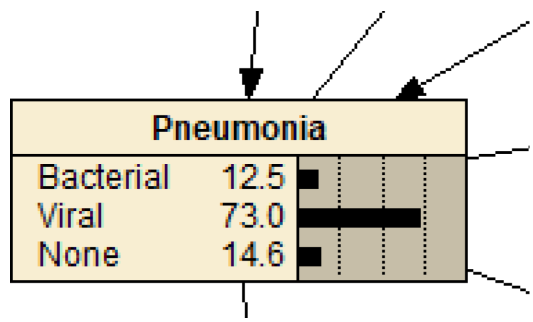
Apply | OK | Reset | Close

Pneumonia	Normal	Consolidation with PE	Consolidation without PE
Bacterial	5	80	15
Viral	68	2	30
None	95	2	3

This shows that there is a small probability of there being bacterial pneumonia if there is consolidation without PE.

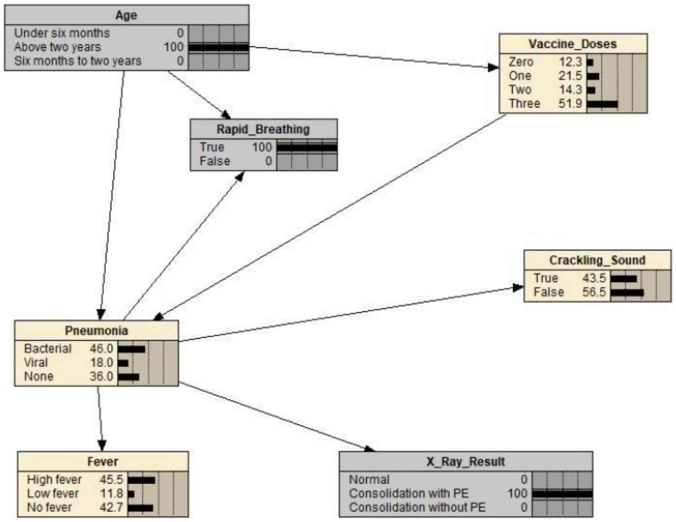
The BN changes like so:

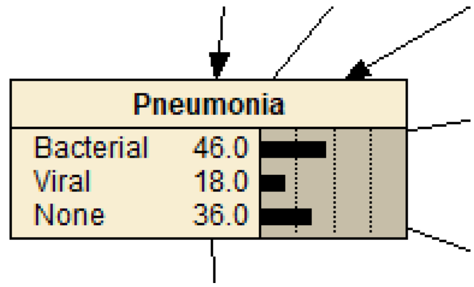




We can see that the probability of the boy having bacterial pneumonia decreases to 12.5%(extremely low probability), while the probability of the boy having viral pneumonia increases to 73%. The probability that the boy has viral pneumonia is now extremely high. This is because of changing the CPT table such that there is an extremely small probability of there being consolidation without PE if there is bacterial pneumonia.

For when the 3 year old boy has rapid breathing and x ray result shows consolidation with PE, using my original parameters:





The boy has a 46% probability of having bacterial pneumonia.

Originally, we can see that the CPT of X_Ray_Result node is like so.

Node: X_Ray_Result Apply OK

Chance % Probability Reset Close

Pneumonia	Normal	Consolidation with PE	Consolidation without PE
Bacterial	5	15	80
Viral	68	2	30
None	95	2	3

Then, I change the CPT value like so:

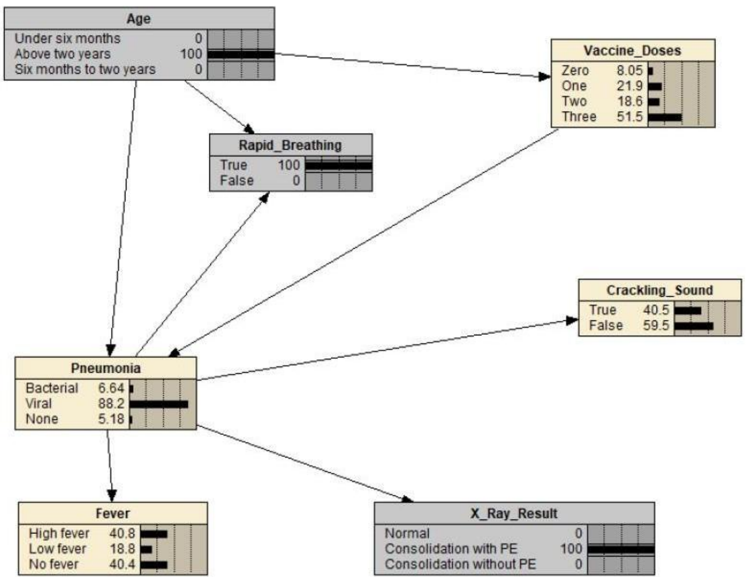
Node: X_Ray_Result Apply OK

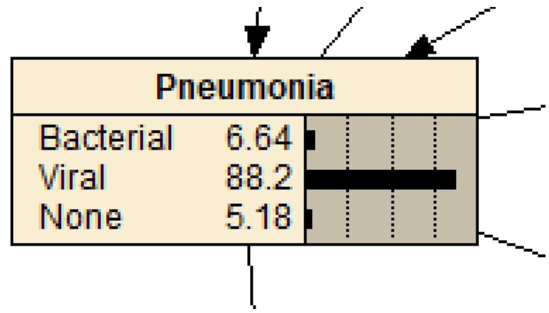
Chance % Probability Reset Close

Pneumonia	Normal	Consolidation with PE	Consolidation without PE
Bacterial	5	15	80
Viral	2	68	30
None	95	2	3

I change the CPT value so that there is a large probability that when there is consolidation with PE, the probability of having viral pneumonia is extremely high.

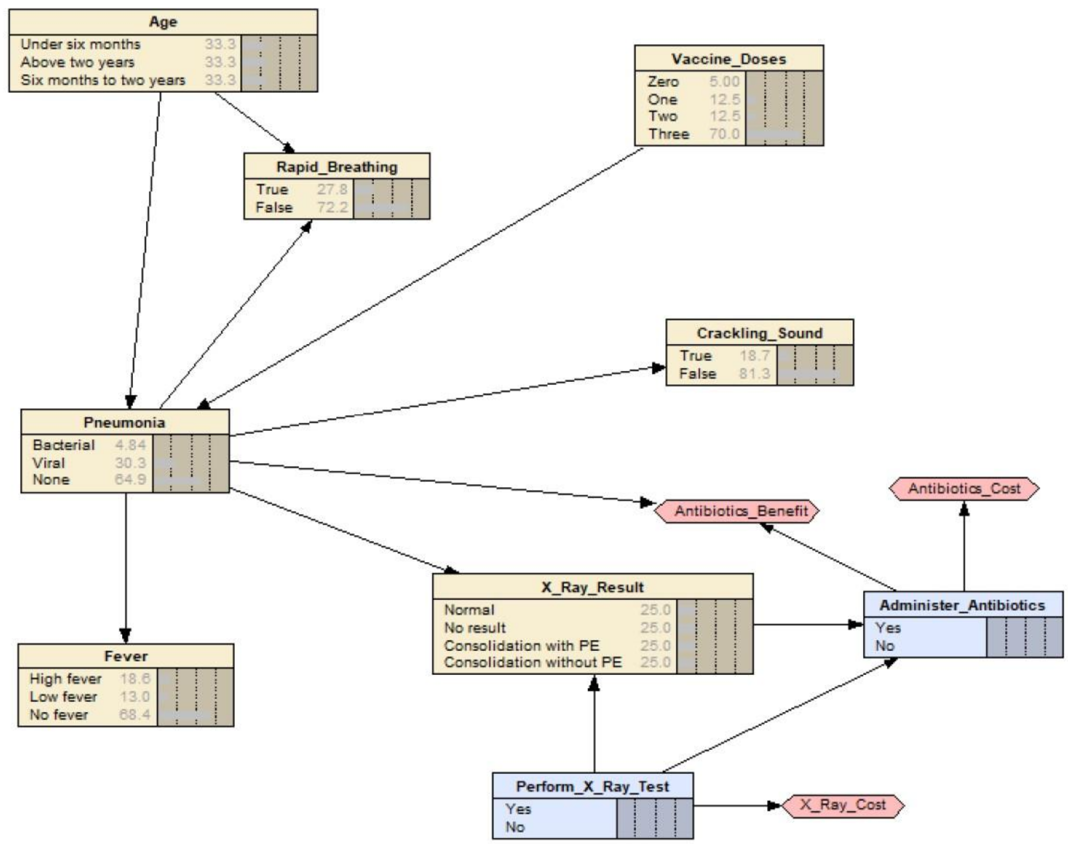
The BN changes like so:





The probability of the boy having bacterial pneumonia is now extremely low(6.64%), whereas the probability of the boy having viral pneumonia is now extremely high(88.2%).This is because, by changing the CPT value so that there is high probability of having viral pneumonia when the x ray result is consolidation without PE, the model now prioritizes viral pneumonia as the more likely cause in this scenario.

My decision network(extended from my simple Bayesian network), adding in decisions on weather to treat with antibiotics and to perform xray or not, and calculating the harms and costs of doing so.



Variable dictionary updated

Node	Description	States	Parents	Relationships To Parents, Justification and source of Parameters and estimates
Pneumonia	Represents the type of pneumonia a child may have: bacterial, viral, or no pneumonia. Is a chance node.	Bacterial, Viral, No Pneumonia	Age, Vaccine Doses	1. Age -> Pneumonia Since younger children are more vulnerable to pneumonia than older children, age has a direct causal relationship with pneumonia. Younger age groups may have weaker immune systems, making them more susceptible to respiratory infections. 2. Vaccine Doses -> Pneumonia Fully vaccinated children are less likely to contract pneumonia, indicating that vaccine doses serve as a protective factor. Thus, vaccine

				doses can be considered a parent variable of pneumonia, as they influence the likelihood of developing the condition. The CPT of pneumonia is built from sources ME and LIT.
Age	Represents the age of the child, which can influence the likelihood of pneumonia. Is a chance node.	Under 6 months, 6 months to 2 years, Above two years	-	Age has no parents in this context because it is a foundational factor. It serves as an independent variable that can influence various health outcomes, including pneumonia. No prior distribution needed since the child's age is known (ME).
Fever	Represents whether the child has a fever, a common sign of bacterial pneumonia. Is a chance node.	High fever, Low fever, No fever	Pneumonia	Pneumonia-> Fever Pneumonia often leads to fever as a response to infection. When the body detects the presence of pathogens causing pneumonia, it activates the immune response, which frequently results in an elevated body temperature (fever). This relationship shows pneumonia is a parent for fever, as having pneumonia increases the likelihood of developing a fever. The likelihood of high or low fever varies with the type of pneumonia. High fever is more common in bacterial pneumonia.(ME), Source of parameters taken from ME.
Rapid Breathing	Represents whether the child is experiencing rapid breathing, a sign of bacterial pneumonia. Is a chance node.	True, False	Pneumonia, Age	1. Pneumonia -> Rapid Breathing Pneumonia can lead to rapid breathing (tachypnea) as the body attempts to compensate for reduced oxygen exchange in the lungs. This relationship indicates that pneumonia is a parent influencing the occurrence of rapid breathing.

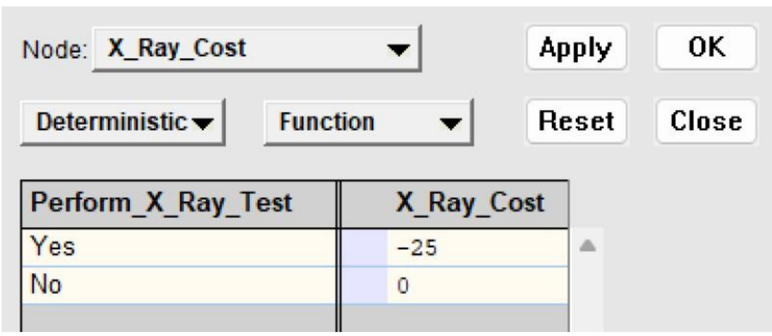
				2. Age -> Rapid Breathing Age can influence the likelihood of rapid breathing, especially in young children and the elderly. Younger children may have a naturally higher respiratory rate compared to older children and adults due to their smaller lung capacity and higher metabolic rates. This is why age is parent of rapid breathing. The type of pneumonia and age affects the likelihood of having rapid breathing(LIT). Source of parameters taken from ME.
Crackling Sound	Represents whether there is a crackling sound in the lungs, often observed with a stethoscope in bacterial pneumonia. Is a chance node.	True, False	Pneumonia	Pneumonia -> Crackling Sound Pneumonia can cause a crackling or "rales" sound in the lungs due to the presence of fluid, mucus, or inflammation in the air sacs (alveoli). This relationship shows that pneumonia is a parent variable that influences the presence of abnormal lung sounds. Crackling sounds are more common with bacterial pneumonia (ME). Rare in viral pneumonia or no pneumonia (ME). Source of parameters: ME.
X-ray Result	Represents the results of an X-ray test, which can help diagnose pneumonia (e.g., consolidation, pleural effusion, or normal). Is a chance node.	Normal, Consolidation with PE, Consolidation without PE, No result	Pneumonia	Pneumonia -> X-Ray result The presence of pneumonia affects the results of an X-ray by leading to characteristic changes in the lung's appearance, such as consolidation, or consolidation with PE. This is why Pneumonia is parent of X-ray result. The type of pneumonia affects likelihood of consolidation appearing on an X-Ray(LIT). Source of parameters: LIT
Vaccine Doses	Represents the number of	Zero, One,	Age	Age-> Vaccine Doses

	pneumococcal vaccine doses the child has received, which protects against bacterial pneumonia. Is a chance node.	Two, Three		Vaccine doses are affected by age.(Eg, if the child is below 6 months, its impossible to be fully vaccinated, my probabilities reflect this). Source of parameters: ME
Perform_X_Ray_Test	Is a decision node. It captures the choice between performing the X-ray test or not. The decision is influenced by its potential outcomes, such as detecting conditions like pneumonia, and the associated costs or risks.	Yes, No	-	Perform_X_Ray_Test does not have any parent node, however it is influenced by utility nodes(X_Ray_Cost, Antibiotics_Benefit) that provide information regarding the cost and benefit of performing the test. This is why it is linked to X_Ray_Cost and Antibiotics_Benefit. There is no parameters in this node.
Administer_Antibiotics	Is a decision node. It captures the choice between administering antibiotics or not. The decision is influenced by its potential outcomes, namely performing x ray test, and the associated costs and risks.	Yes, No	X_Ray_Result, Perform_X_Ray_Test	1. X_Ray_Result-> Administer_Antibiotics The result of the x ray will influence the decision of weather or not to administer antibiotics. Justification: Ex: if the x ray results are consolidation with PE, there is high chance that the person has bacterial pneumonia, which then administer antibiotics will have high expected utility to administer antibiotics. 2. Perform_X_Ray_Test -> Administer_Antibiotics The decision of performing x ray test or not will influence the decision of weather or not to administer antibiotics. Justification: Ex: if the decision to not perform an x ray test is made, this means that there is high possibility that the person has no pneumonia.

				So, there is no reason to administer antibiotics.
X_Ray_Cost	Is a utility node. The utility table stores values about the costs and risks associating to performing the x ray or not.	-	Perform_X_Ray_Test	Perform_X_Ray_Test -> X_Ray_Cost The decision to perform the x ray is influenced by the cost of x ray, since x rays are harmful due to the exposure to radiation, so this link helps weigh the benefits and risks of doing x ray. Justification: Ex: if the cost of performing the x ray is deemed reasonable and justified based on the expected outcomes, a healthcare provider may be more inclined to proceed with the test.
Antibiotics_Cost	Is a utility node. The utility table stores values about the costs and risks associating with administering the antibiotics or not.	-	Administer_Antibiotics	Administer_Antibiotics -> Antibiotics_Cost The decision to administer antibiotics is influenced by the cost of antibiotics, since misusing antibiotics is harmful to the body and can lead to bacterial resistance. Justification: Ex: if the potential costs of administering antibiotics do not outweigh the other benefits, a healthcare provider may opt for alternative treatments or wait for further diagnostic confirmation before proceeding.
Antibiotics_Benefit	Is a utility node. The utility table stores values about the costs and risks of giving antibiotics or not in different pneumonia types.	-	Administer_Antibiotics, Pneumonia	1. Administer_Antibiotics -> Antibiotics_Benefit The decision to administer antibiotics is influenced by the benefits of antibiotics. Justification: Ex: if the person is diagnosed as not having pneumonia, the benefit of administering antibiotics is negative value, since this is considered misusing antibiotics and is harmful to the body. 2. Pneumonia -> Antibiotics_Benefit

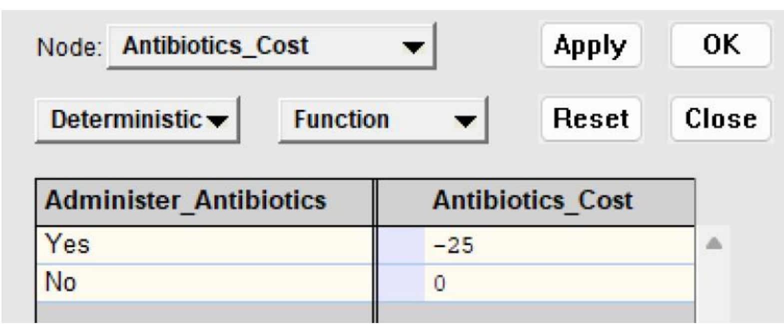
				The decision to administer antibiotics is influenced by the pneumonia type, Justification: Ex: If the person is diagnosed with having bacterial pneumonia, the benefit of treating with antibiotics is high, since this is the correct way to treat bacterial pneumonia.

Justifying my new utilities table:



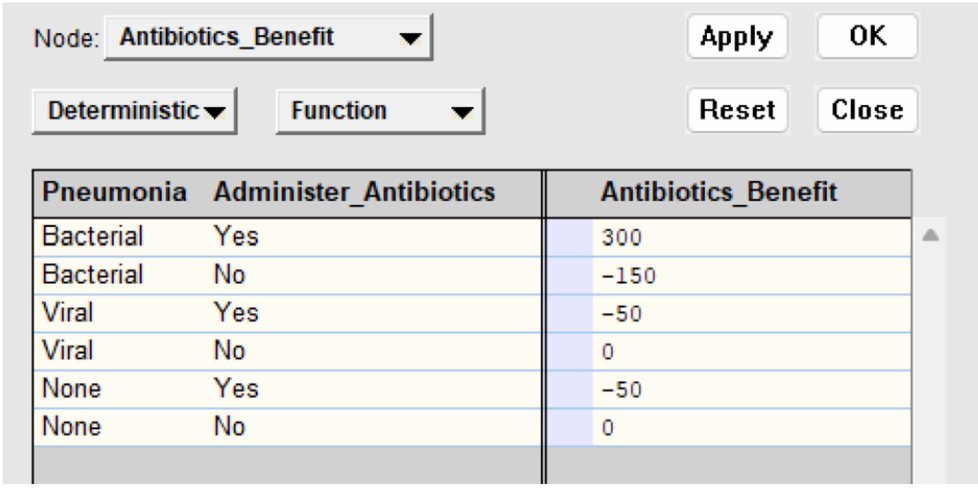
Perform_X_Ray_Test	X_Ray_Cost
Yes	-25
No	0

The x ray cost for performing an x ray test is -25, since performing x rays are harmful to the body, since they involve exposure to radiation. Whereas, not performing x rays does nothing to the body, so there is 0 cost.



Administer_Antibiotics	Antibiotics_Cost
Yes	-25
No	0

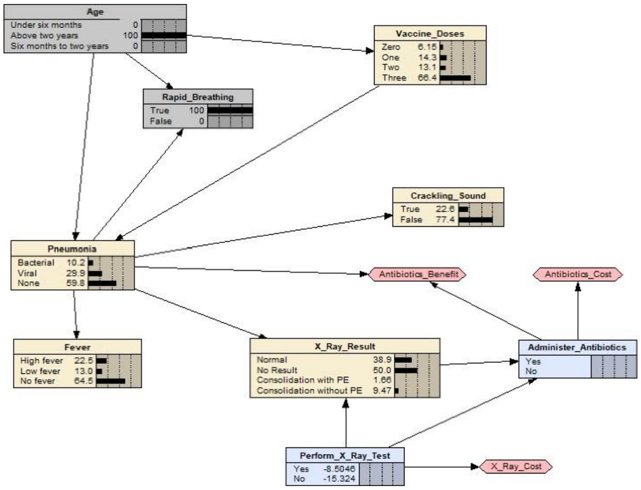
The cost of administering antibiotics is -25, since misuse of antibiotics can lead to antibiotic resistance. Whereas, not administering antibiotics does nothing to the body, so the cost is 0.

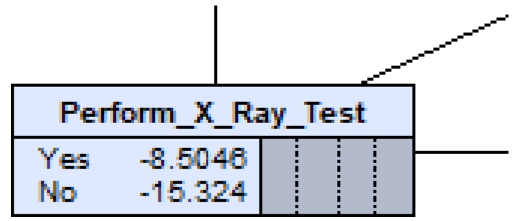


Pneumonia	Administer_Antibiotics	Antibiotics_Benefit
Bacterial	Yes	300
Bacterial	No	-150
Viral	Yes	-50
Viral	No	0
None	Yes	-50
None	No	0

When there is bacterial pneumonia, and antibiotics are administered , the antibiotics benefit is 300, since the pneumonia is accurately treated. If its bacterial pneumonia but no antibiotics are administered, the utility value is -150, since pneumonia is untreated. If its viral pneumonia and antibiotics are administered, utility value -50, since antibiotics are only supposed to cure bacterial pneumonia. If its viral pneumonia and no antibiotics are administered, utility value is 0, since there is no benefit from it. If there is no pneumonia, but antibiotics are administered, utility value is -50, since antibiotics are inappropriately used. Lastly, if there is no pneumonia, and antibiotics are not administered, utility value is 0, since there is no benefit from it.

If the boy from the previous example did an xray:



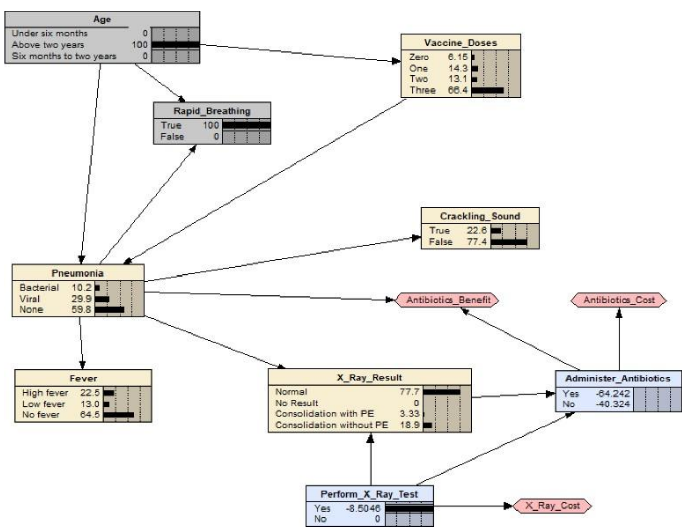


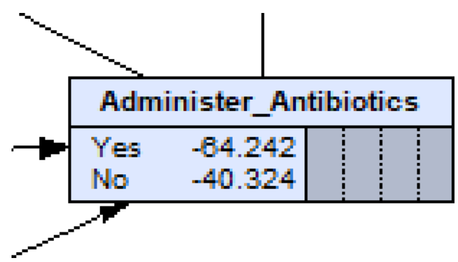
3 year old boy with rapid breathing

Expected utilities:

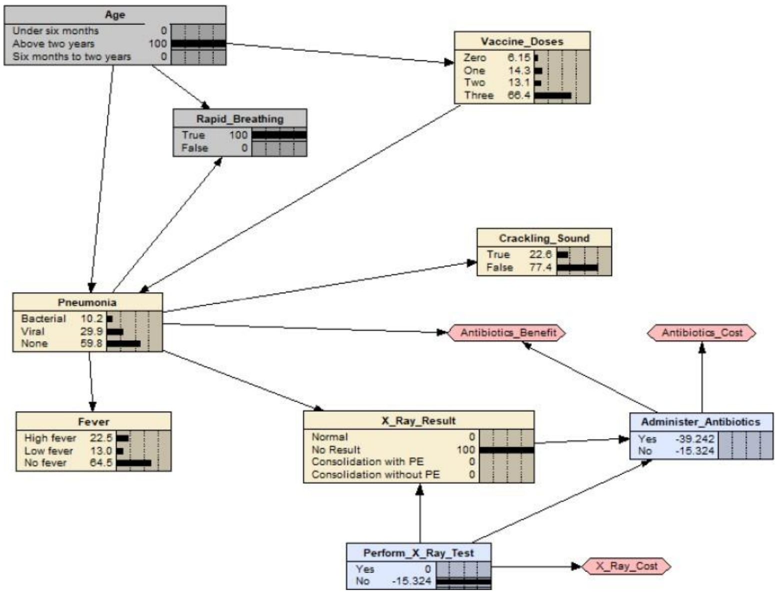
Perform_X_Ray_Test: The expected utility for performing the X-ray test (Yes) is-8.5048, while the expected utility for not performing the X-ray test (No) is-15.324. The expected utility for performing the X-ray test is higher than the expected utility for not performing the X-ray test.

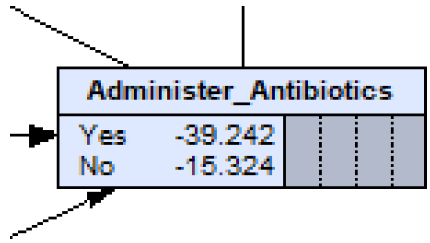
Administer_Antibiotics:





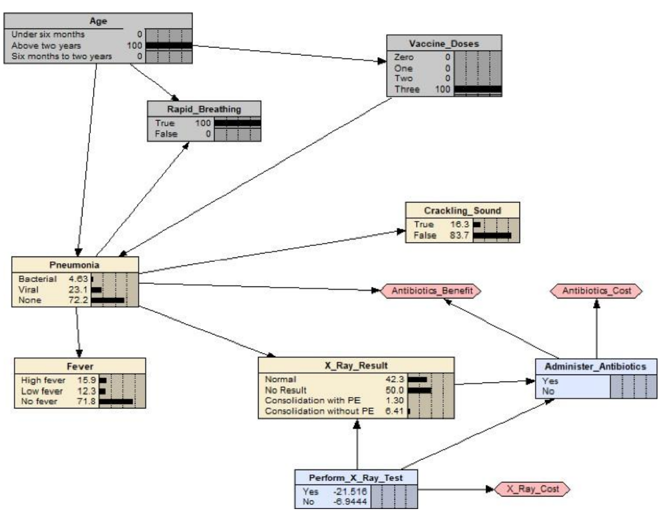
If choose to perform x ray test, the expected utility for administering antibiotics(Yes) is -64.242, while the expected utility for not administering antibiotics(No) is -40.324. The expected utility for administering antibiotics is lower than expected utility for not administering antibiotics.

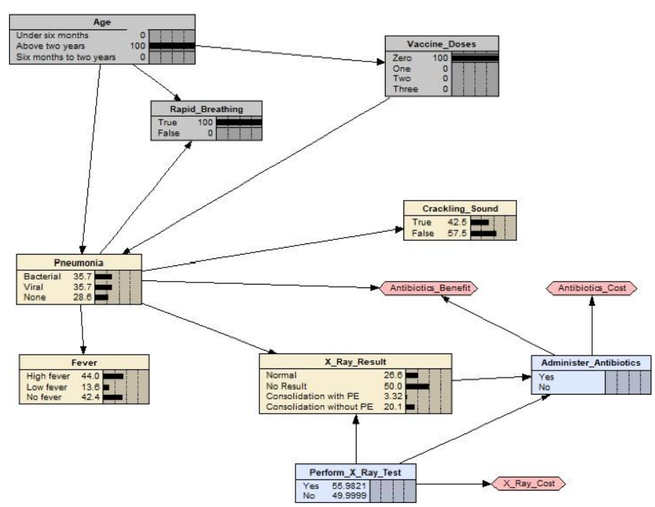




If we do choose to not perform x ray test, the expected utility for administering antibiotics(Yes) is -39.242, while the expected utility for not administering antibiotics(No) is -15.324. The expected utility for administering antibiotics is lower than expected utility for not administering antibiotics.

If the boy has had all his vaccination doses vs has had none vaccination:





3 year old boy with rapid breathing

and 3 vaccination doses

3 year old boy with rapid breathing

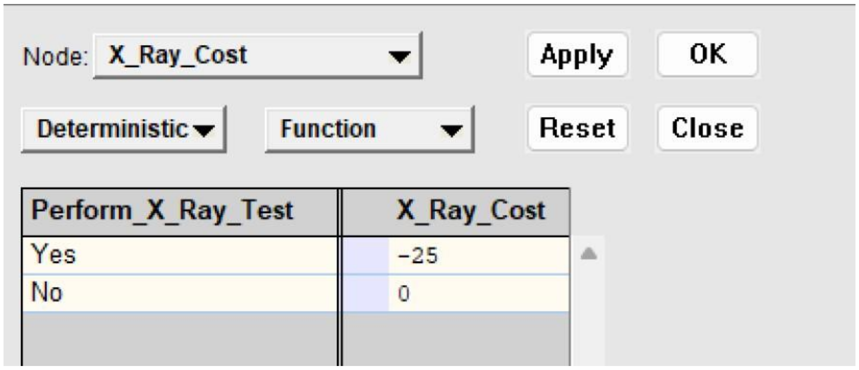
and no vaccination doses

If the boy has received all his vaccination doses, the expected utility for Perform_X_Ray_Test (Do_Xray) is higher for No (-6.9444) and lower for Yes (-21.516).

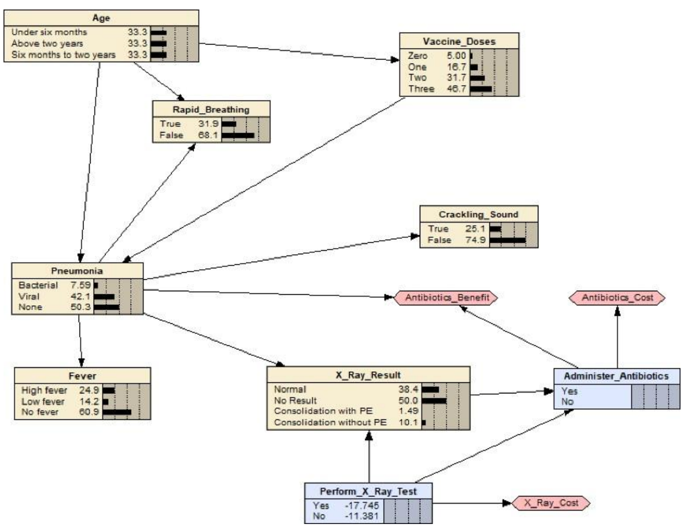
On the other hand, if the boy has not received any vaccination doses, the expected utility for Perform_X_Ray_Test (DoXray) is higher for Yes (55.9821) and lower for No (49.9999).

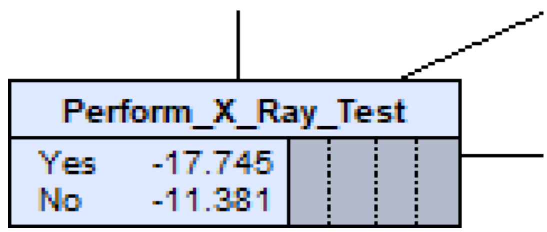
This means that if the boy has completed all the vaccination doses, the probability that he should do an x-ray decreases. The expected utility for DoXray decreases because the risk of severe bacterial pneumonia is lower, reducing the need for an X-ray. In this case, it may not be worth doing the X-ray because the boy is less likely to have bacterial pneumonia, and the likelihood of a false positive increases, and the harm and cost of the X-ray might outweigh the potential benefit.. On the other hand, if the boy has had no vaccine doses, the risk of bacterial pneumonia increases, and the probability of detecting consolidation or pleural effusion through an X-ray increases, making the expected utility of doing an X-ray higher. In this scenario, it would be worth doing the X-ray, as the potential benefit of detecting a serious illness is much greater.

The sensitivity of my doxray decision to my utilities table:

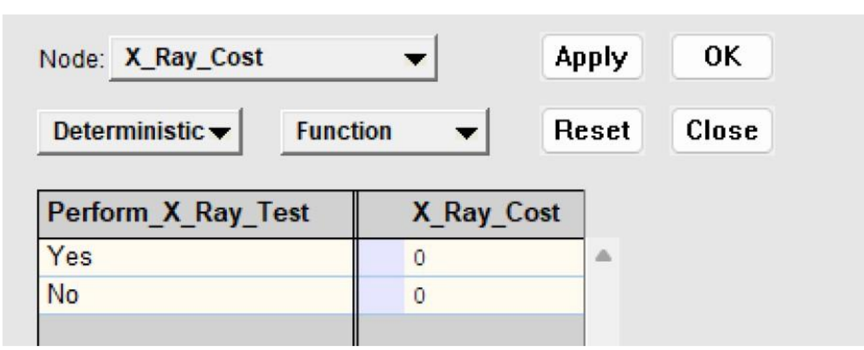


In my original implementation, my X-Ray Cost utility table is set like so: if perform x ray test is yes, the cost is -10, as performing x ray is harmful to the patient, whereas if perform x ray test is no, the cost is 0, since there is no harm or benefit in not doing x ray. This is my Perform_X_Ray_Test decision values:

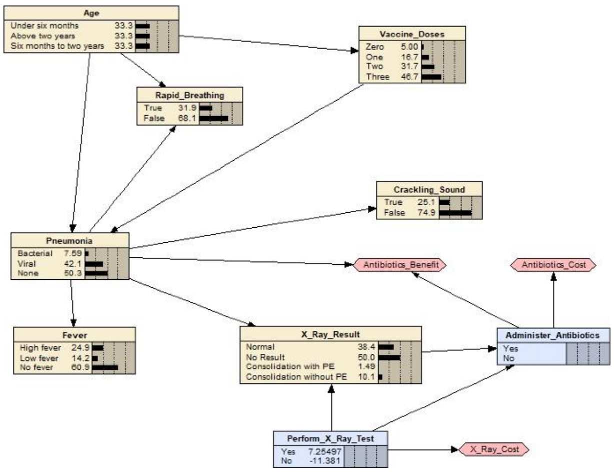


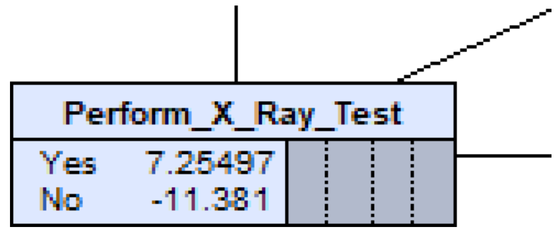


Now, I change the value in the X_Ray_Cost utility table so that there is no harm in performing x ray test.



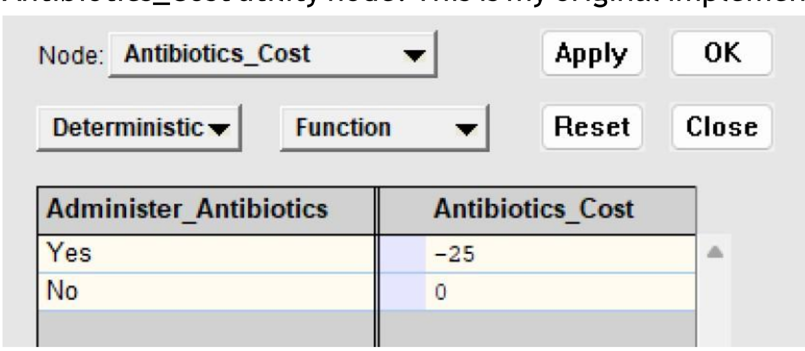
When this happens, my Perform_X_Ray_Test decision values changes to:





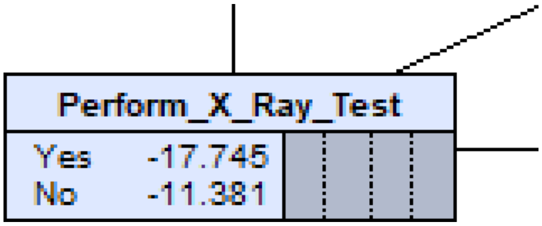
From this, I can see that for my expected utilities, the expected utility for do x ray has increased(from -17.745 to 7.25497) whereas the expected utility for don't do x ray has not changed(-11.381).However, in the original values, the expected utility for do x ray is lower than don't do x ray, whereas after I changed the values, the expected utility for do x ray is higher than don't do x ray. This is because of changing the values in the utility table, by changing the x ray cost of performing the x ray from -25 to 0(means performing x ray causes no harm), the Do Xray decision node(Perform_X_Ray_Test) will lean towards performing x ray test.

Then, I reset my X_Ray_Cost utility node to my original implementation, and look at my Antibiotics_Cost utility node. This is my original implementation:

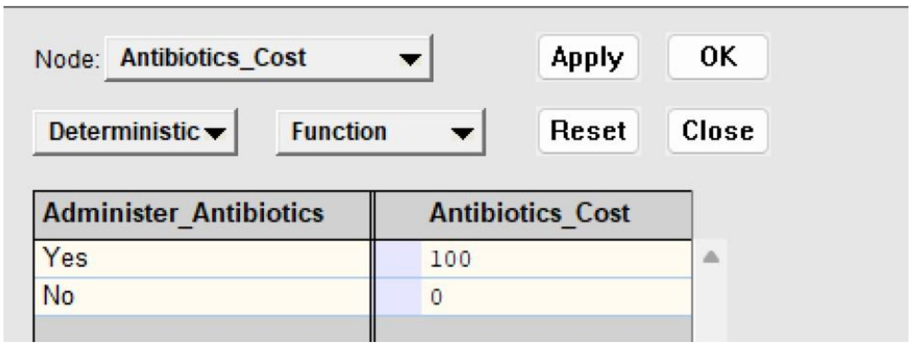


In my original implementation, I set values of antibiotics cost when antibiotics are administered to -25, and antibiotics cost to 0 when no antibiotics are administered.

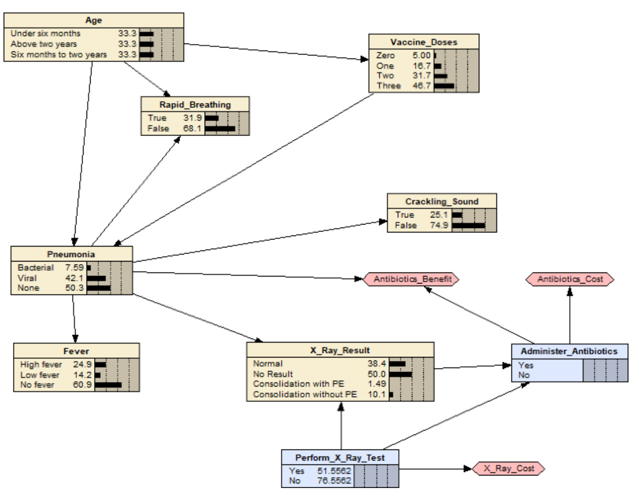
This is my Do_Xray (Perform_X_Ray_Test) expected utility values:

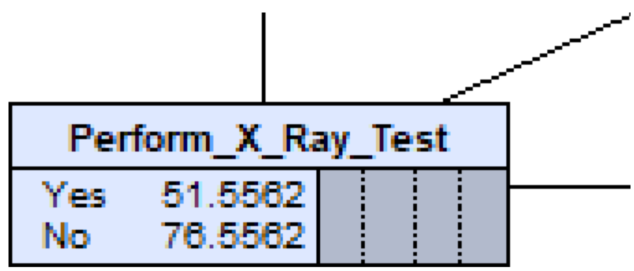


Now, I change my values in my Antibiotics_Cost utility table like so:



As compared to my original implementation, I make the cost of administering antibiotics a lot more favourable than not administering antibiotics. When this is applied, my Perform_X_Ray_Test decision values changes to:



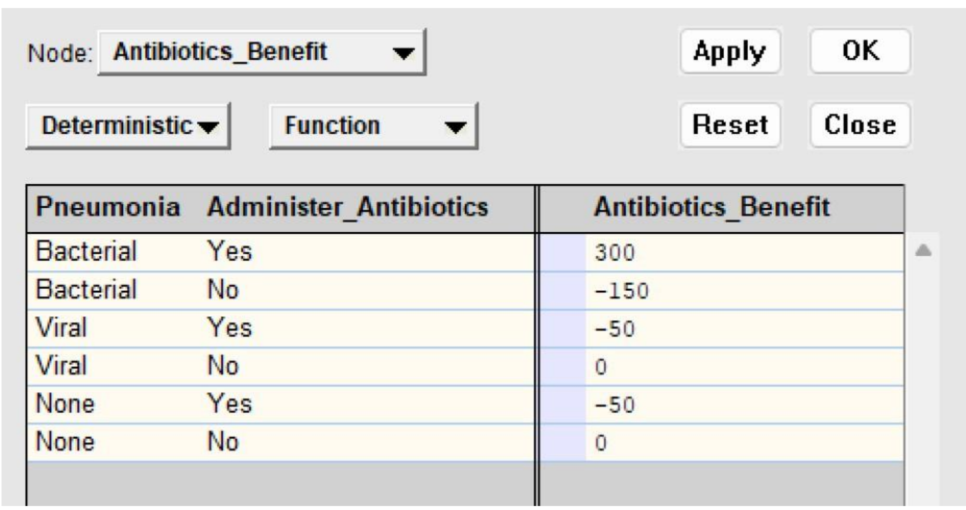


From here, we can see that the expected utility value of Perform_X_Ray_Test changes from expected utility of yes being -17.745 to 51.5582 and expected utility of no being -11.381 to

76.5582. This is because, when we change the utility values(administering antibiotics utility = 100), this makes administering antibiotics highly favorable. The model will prioritize actions that lead to administering antibiotics whenever possible, as this decision will contribute substantially to the overall utility. On the other hand, by deciding to not change the utility value for not administering antibiotics(keep at 0), means that this option remains far less favorable than administering antibiotics. The gap between the two options (100 vs. 20) will drive the model to heavily favor the decision to administer antibiotics.

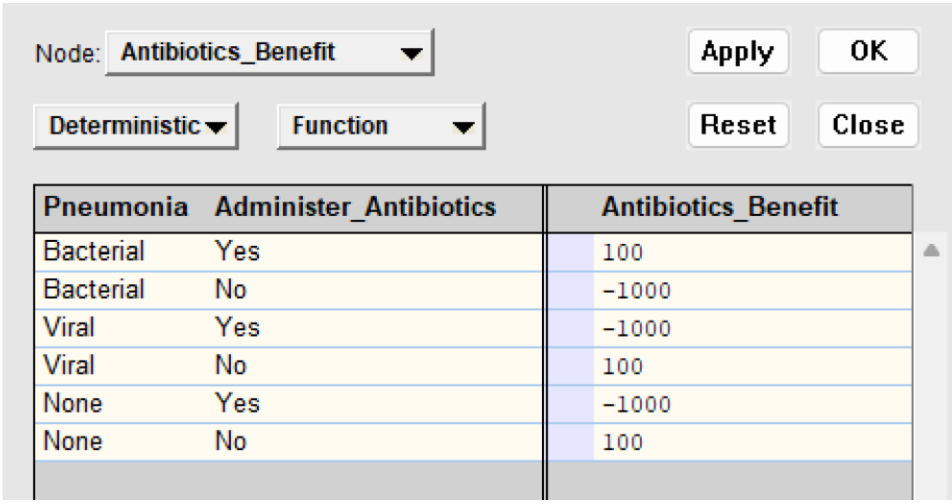
Increasing the utility for administering antibiotics from -25 to 100 makes it the strongly preferred action, which increases the expected utility for both performing and not performing an X-ray, since the model recognizes that whether an X-ray is performed or not, administering antibiotics will result in a highly favorable outcome, so the expected utility for both options increases.

Lastly, we look at the Antibiotics Benefit utility node. Initially, these were my utility values:



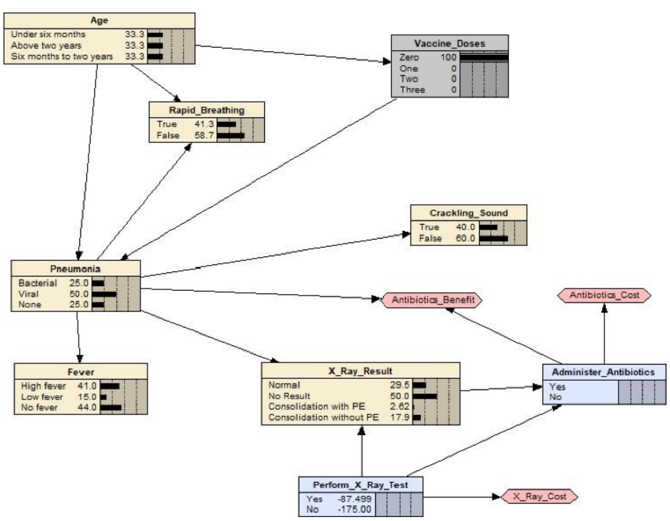
Pneumonia	Administer_Antibiotics	Antibiotics_Benefit
Bacterial	Yes	300
Bacterial	No	-150
Viral	Yes	-50
Viral	No	0
None	Yes	-50
None	No	0

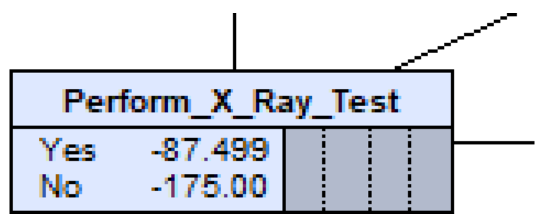
Then I change it like so:



Pneumonia	Administer_Antibiotics	Antibiotics_Benefit
Bacterial	Yes	100
Bacterial	No	-1000
Viral	Yes	-1000
Viral	No	100
None	Yes	-1000
None	No	100

I change the values in the utility Antibiotics Benefit table, so that there is an extremely great penalty when misdiagnosed(example: when it is bacterial pneumonia and antibiotics is not administered, there is extreme penalty of -1000). This makes my expected utilities of my Perform_X_Ray_Test change like so:





From here, we can see that the expected utility value of Perform_X_Ray_Test changes from expected utility of yes being -17.745 to -87.499 and expected utility of no being -11.381 to -175.00. This is because, the significant penalties associated with misdiagnosing patients (administering antibiotics when unnecessary or failing to administer them when bacterial pneumonia is present) lead to a drastic drop in expected utility. This makes the potential consequences of incorrect decisions much more pronounced. With the higher penalties associated with misdiagnosis, the utility of performing the X-ray rises significantly. The test helps prevent the costly consequences of incorrectly diagnosing a condition, making it a more attractive option. This is why the expected utility of performing an x ray test becomes higher than expected utility of not performing the x ray test.